Regression Discontinuity

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I. Introduction

This chapter introduces *regression discontinuity* (RD) designs, which exploit a discontinuity in the probability of treatment at a known cutoff value of a running variable. Identification is *local* to the cutoff: under smoothness of potential outcomes in the running variable, units just below and just above the threshold are comparable. We distinguish *sharp* designs (perfect compliance) and *fuzzy* designs (imperfect compliance), and discuss how the interpretation of the RD estimand depends on additional local assumptions (ATE at the cutoff vs. LATE for compliers at the cutoff).

II. The fundamental RD assumption

In what we have seen so far, the main assumption in the matching context is conditional independence, $(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i|X_i$, whereas in the IV context we assume orthogonality and relevance of the instrument, $(Y_{1i}, Y_{0i}) \perp\!\!\!\perp Z_i|X_i$ and $D_i \not\perp\!\!\!\perp Z_i|X_i$ respectively. The relevance condition can also be expressed as $P(D_i = 1|Z_i = z) \neq P(D_i = 1|Z_i = z')$ for some $z \neq z'$. In ***regression discontinuity*** (RD) we consider a situation where there is a continuous variable Z_i that is not necessarily a valid instrument *globally* (it need not satisfy an exogeneity assumption away from the cutoff), but such that treatment assignment is a discontinuous function of Z_i at a known cutoff z_0. Identification relies on an asymmetry: there is a *discontinuity* in the dependence of D_i on Z_i at z_0, but *continuity* in the dependence of potential outcomes (Y_{1i}, Y_{0i}) on Z_i at z_0. Intuitively, agents cannot precisely manipulate their position around z_0, so that units just below and just above the cutoff are comparable. RD methods have much potential in economic applications because geographic boundaries or program rules (e.g. eligibility thresholds) often create usable discontinuities.

More formally, discontinuity in treatment assignment but continuity in potential outcomes means that there is at least a known value $z = z_0$ such that:

$$\lim_{z \to z_0^+} P(D_i = 1|Z_i = z) \neq \lim_{z \to z_0^-} P(D_i = 1|Z_i = z) \tag{1}$$

$$\lim_{z \to z_0^+} P(Y_{ji} \leq r|Z_i = z) = \lim_{z \to z_0^-} P(Y_{ji} \leq r|Z_i = z) \quad (j = 0, 1) \tag{2}$$

The first condition in (1) is a *relevance* condition, while (2) is a *local orthogonality/continuity* condition on potential outcomes at the cutoff.

Implicit regularity conditions are: (i) the existence of the limits, and (ii) that Z_i has positive density in a neighborhood of z_0. We abstract from conditioning covariates for the time being for simplicity.

Early RD literature in Psychology (e.g. Cook and Campbell, 1979) distinguishes between ***sharp*** and ***fuzzy*** designs depending on the nature of compliance with the assignment rule. In a **sharp** design, treatment is a deterministic function of the running variable:

$$D_i = \mathbb{1}\{Z_i \geq z_0\}, \tag{3}$$

so that:

$$\begin{aligned} \lim_{z \to z_0^+} \mathbb{E}[D_i|Z_i = z] = 1, \\ \lim_{z \to z_0^-} \mathbb{E}[D_i|Z_i = z] = 0. \end{aligned} \tag{4}$$

In a **fuzzy** design, compliance is imperfect and the probability of treatment changes discontinuously at z_0:

$$P(D_i = 1|Z_i = z_0 - \varepsilon) \neq P(D_i = 1|Z_i = z_0 + \varepsilon),$$

for $\varepsilon > 0$ small. A sharp RD is a special case of a fuzzy RD with *perfect compliance*.

III. Homogeneous Treatment Effects

Like in the IV setting, the case of homogeneous treatment effects is useful to present the basic RD estimator. Suppose that $\alpha = Y_{1i} - Y_{0i}$ is constant, so that:

$$Y_i = \alpha D_i + Y_{0i} \tag{5}$$

Taking conditional expectations given $Z_i = z$ and left-side and right-side limits:

$$\begin{aligned} \lim_{z \to z_0^+} \mathbb{E}[Y_i|Z_i = z] = \alpha \lim_{z \to z_0^+} \mathbb{E}[D_i|Z_i = z] + \lim_{z \to z_0^+} \mathbb{E}[Y_{0i}|Z_i = z] \\ \lim_{z \to z_0^-} \mathbb{E}[Y_i|Z_i = z] = \alpha \lim_{z \to z_0^-} \mathbb{E}[D_i|Z_i = z] + \lim_{z \to z_0^-} \mathbb{E}[Y_{0i}|Z_i = z], \end{aligned} \tag{6}$$

which leads to the consideration of the following RD parameter:

$$\alpha = \frac{\lim_{z \to z_0^+} \mathbb{E}[Y_i|Z_i = z] - \lim_{z \to z_0^-} \mathbb{E}[Y_i|Z_i = z]}{\lim_{z \to z_0^+} \mathbb{E}[D_i|Z_i = z] - \lim_{z \to z_0^-} \mathbb{E}[D_i|Z_i = z]}, \tag{7}$$

which is determined provided the relevance condition in Equation (1) is satisfied, and equals α provided the independence condition in Equation (2) holds. The

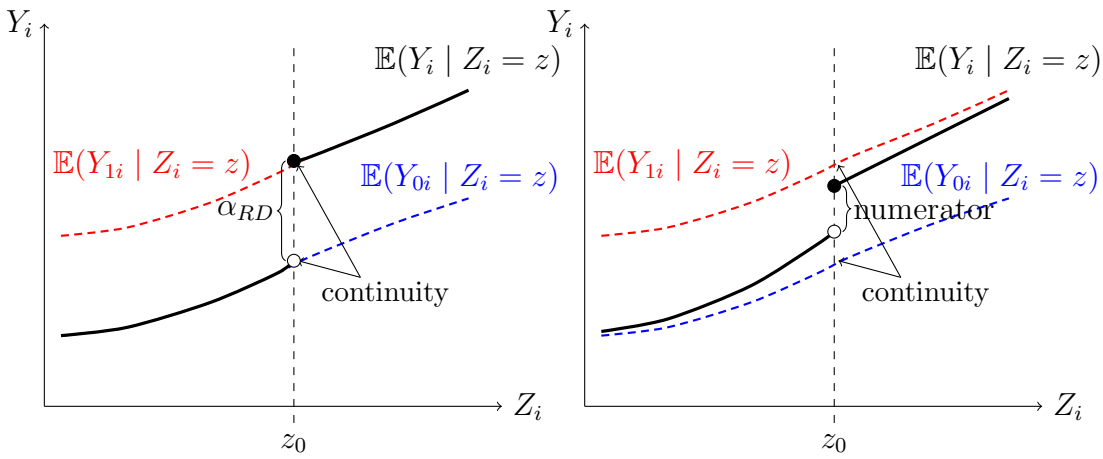
numerator is the *reduced form* discontinuity in outcomes, while the denominator is the *first stage* discontinuity in treatment. In fuzzy designs, the ratio is therefore a local *Wald estimand*.

In the case of a sharp design, the denominator is unity so that:

$$\alpha = \lim_{z \to z_0^+} \mathbb{E}[Y_i | Z_i = z] - \lim_{z \to z_0^-} \mathbb{E}[Y_i | Z_i = z], \tag{8}$$

which can be regarded as a matching-type situation, in the same way that the general case can be regarded as an IV-type situation. So the basic idea is to obtain a treatment effect by comparing the average outcome left of the discontinuity with the average outcome to the right of discontinuity, relative to the difference between the left and right propensity scores. Intuitively, considering units within a small interval around the cutoff point is similar to a randomized experiment at the cutoff point.

To grasp some intuition about what these estimands identify, the two figures below illustrate the behavior of conditional expectations around the cutoff z_0 in sharp and fuzzy regression discontinuity designs. In both cases, the key identifying assumption is continuity of the potential outcome means $\mathbb{E}[Y_{0i} \mid Z_i = z]$ and $\mathbb{E}[Y_{1i} \mid Z_i = z]$ at z_0, which follows from the idea that agents cannot precisely manipulate their position around the threshold.



Sharp regression discontinuity

Fuzzy regression discontinuity

In the sharp design, treatment is a deterministic function of the running variable, so that $D_i = \mathbb{1}\{Z_i \geq z_0\}$. As a consequence, for values of Z_i just below the cutoff we observe $Y_i = Y_{0i}$, while for values just above the cutoff we observe $Y_i = Y_{1i}$. Under continuity of the potential outcome means, the observed conditional expectation $\mathbb{E}[Y_i \mid Z_i = z]$ thus coincides with $\mathbb{E}[Y_{0i} \mid Z_i = z]$ to the left of z_0 and with

$\mathbb{E}[Y_{1i} \mid Z_i = z]$ to the right. The jump in $\mathbb{E}[Y_i \mid Z_i = z]$ at the cutoff is directly attributable to treatment and identifies the causal effect at the threshold, α_{RD}.

In the fuzzy design, compliance with the assignment rule is imperfect and treated and untreated units coexist on both sides of the cutoff. As a result, the observed conditional mean $\mathbb{E}[Y_i \mid Z_i = z]$ is a mixture of potential outcomes, and its discontinuity at z_0 is not itself a treatment effect. Instead, the jump in outcomes reflects both the causal effect of treatment and the change in the probability of receiving treatment at the cutoff. Identification therefore requires rescaling the discontinuity in $\mathbb{E}[Y_i \mid Z_i = z]$ by the corresponding discontinuity in $\mathbb{E}[D_i \mid Z_i = z]$. The resulting Wald ratio defines the RD estimand, which can be interpreted as an average treatment effect at the cutoff under local conditional independence, or as a local average treatment effect for compliers at the cutoff under a local monotonicity assumption.

IV. Heterogeneous Treatment Effects

Now suppose that:

$$Y_i = \alpha_i D_i + Y_{0i}. \tag{9}$$

It is useful again to distinguish sharp and fuzzy designs.

A. *Sharp design*

In the sharp design, since $D_i = \mathbb{1}\{Z_i \geq z_0\}$ we have:

$$\mathbb{E}[Y_i|Z_i = z] = \mathbb{E}[\alpha_i|Z_i = z]\, \mathbb{1}\{z \geq z_0\} + \mathbb{E}[Y_{0i}|Z_i = z]. \tag{10}$$

In other words, conditioning on a value z for Z_i, individuals are treated if $z \geq z_0$, and thus we observe $Y_i = Y_{1i} = \alpha_i + Y_{0i}$, and untreated if $z \leq z_0$, in which case we observe $Y_i = Y_{0i}$. Thus, to obtain an average treatment effect for individuals at the threshold value z_0, that is, α_{RD} defined as:

$$\alpha_{RD} \equiv \mathbb{E}[\alpha_i|Z_i = z_0], \tag{11}$$

we rewrite (10) as:

$$\begin{aligned}
\mathbb{E}[Y_i|Z_i = z] &= \mathbb{E}[\alpha_i|Z_i = z]\, \mathbb{1}\{z \geq z_0\} + \mathbb{E}[Y_{0i}|Z_i = z] \pm \mathbb{E}[\alpha_i|Z_i = z_0]\, \mathbb{1}\{z \geq z_0\} \\
&= \alpha_{RD}\, \mathbb{1}\{z \geq z_0\} + \mathbb{E}[Y_{0i}|Z_i = z] \\
&\qquad\qquad + (\mathbb{E}[\alpha_i|Z_i = z] - \mathbb{E}[\alpha_i|Z_i = z_0])\, \mathbb{1}\{z \geq z_0\} \\
&\equiv \alpha_{RD} D_i + k_{z_0}(z).
\end{aligned} \tag{12}$$

In a sharp design, treatment is fully determined by the observed running variable Z_i, so this is a *selection on observables* setting: conditional on Z_i, treatment status is not driven by unobservables. The role of the control function $k_{z_0}(z)$ is to absorb smooth differences in outcomes related to Z_i (including smooth variation in $\mathbb{E}[Y_{0i}|Z_i = z]$ and smooth variation in $\mathbb{E}[\alpha_i|Z_i = z]$ away from z_0); it is not itself a causal object. Therefore, the OLS population coefficient on D_i in the equation:

$$Y_i = \alpha_{RD} D_i + k_{z_0}(Z_i) + v_i \tag{13}$$

equals $\mathbb{E}[\alpha_i|Z_i = z_0]$, which is the causal effect of interest (an average treatment effect for individuals with Z_i right below or above the discontinuity).

The control function $k_{z_0}(z)$ is nonparametrically identified (e.g. including a high-order polynomial in Z_i —or $Z_i - z_0$— in the OLS regression interacted with a dummy $\mathbb{1}\{Z_i \geq z_0\}$). Note that if the treatment effect is homogeneous, $k(z)$ coincides with $\mathbb{E}[Y_{0i}|Z_i = z]$, but not in general.

B. Fuzzy design

In the fuzzy design, D_i not only depends on $\mathbb{1}\{Z_i \geq z_0\}$, but also on other unobserved variables. Thus, D_i is an endogenous variable in Equation (13). However, we can still use $\mathbb{1}\{Z_i \geq z_0\}$ as an instrument for D_i in such equation to identify α_{RD}, at least in the homogeneous case. The connection between the fuzzy design and the instrumental variables perspective was first made explicit in van der Klaauw (2002).

The assumptions deliver different causal parameters: an *ATE at the cutoff* under local conditional independence, or a *LATE for compliers at the cutoff* under local monotonicity. Consider first the weak conditional independence assumption:

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i|Z_i = z \quad \text{for } z \text{ near } z_0, \tag{14}$$

that is, for $z = z_0 \pm e$, where e is an arbitrarily small positive number, or simply:

$$F(Y_{ji}|D_i = 1, Z_i = z_0 \pm e) = F(Y_{ji}|Z_i = z_0 \pm e) \quad (j = 0, 1). \tag{15}$$

Thus, we are assuming that treatment assignment is exogenous in the neighborhood of z_0. An implication is:

$$\mathbb{E}[\alpha_i D_i|Z_i = z_0 \pm e] = \mathbb{E}[\alpha_i|Z_i = z_0 \pm e]\, \mathbb{E}[D_i|Z_i = z_0 \pm e]. \tag{16}$$

Proceeding as before, we have:

$$\begin{aligned}\lim_{z\to z_0^+}\mathbb{E}[Y_i|Z_i=z] &= \lim_{z\to z_0^+}\mathbb{E}[\alpha_i|Z_i=z]\,\mathbb{E}[D_i|Z_i=z] + \lim_{z\to z_0^+}\mathbb{E}[Y_{0i}|Z_i=z]\\ \lim_{z\to z_0^-}\mathbb{E}[Y_i|Z_i=z] &= \lim_{z\to z_0^-}\mathbb{E}[\alpha_i|Z_i=z]\,\mathbb{E}[D_i|Z_i=z] + \lim_{z\to z_0^-}\mathbb{E}[Y_{0i}|Z_i=z].\end{aligned} \tag{17}$$

Noting that $\lim_{z\to z_0^+}\mathbb{E}[\alpha_i|Z_i=z] = \lim_{z\to z_0^-}\mathbb{E}[\alpha_i|Z_i=z] = \alpha_{RD}$, subtracting one equation from the other, and rearranging the terms we obtain:

$$\begin{aligned}\alpha_{RD} &\equiv \mathbb{E}[Y_{1i}-Y_{0i}|Z_i=z_0]\\ &= \frac{\lim_{z\to z_0^+}\mathbb{E}[Y_i|Z_i=z] - \lim_{z\to z_0^-}\mathbb{E}[Y_i|Z_i=z]}{\lim_{z\to z_0^+}\mathbb{E}[D_i|Z_i=z] - \lim_{z\to z_0^-}\mathbb{E}[D_i|Z_i=z]}.\end{aligned} \tag{18}$$

That is, the RD parameter can be interpreted as the average treatment effect at z_0.

Hahn, Todd, and van der Klaaw (2001) also consider an alternative LATE-type of assumption. Let D_{zi} be the potential assignment indicator associated with $Z_i = z$, and for some $\bar{\varepsilon} > 0$ and any pair $(z_0-\varepsilon, z_0+\varepsilon)$ with $0<\varepsilon<\bar{\varepsilon}$ suppose the local monotonicity assumption:

$$D_{z_0+\varepsilon,i} \geq D_{z_0-\varepsilon,i} \text{ for all units } i \text{ in the population.} \tag{19}$$

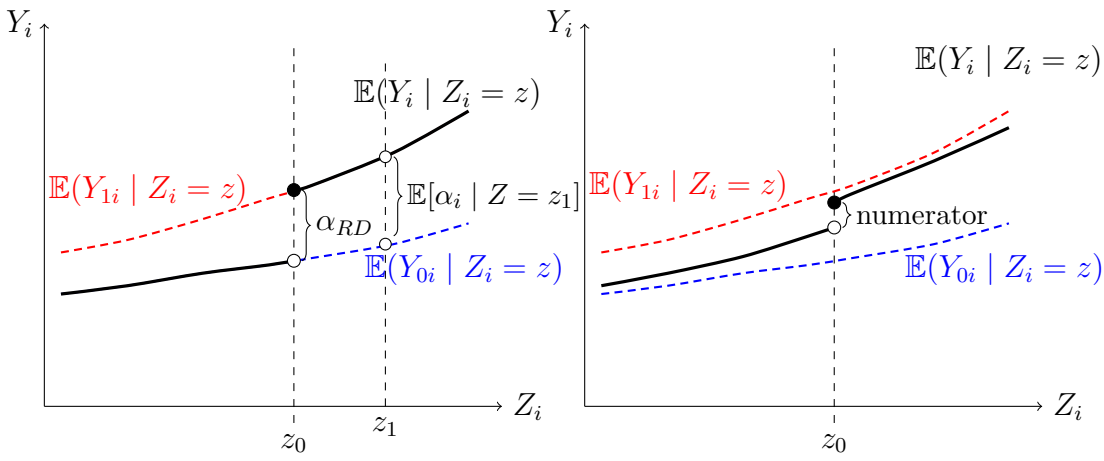
Sometimes, the local conditional independence assumption could be problematic, especially in fuzzy designs, but the monotonicity assumption is not. In such case, it can be shown that α_{RD} identifies the local average treatment effect at $z = z_0$:

$$\alpha_{RD} = \lim_{\varepsilon\to 0^+}\mathbb{E}[Y_1 - Y_0|D_{z_0+\varepsilon} - D_{z_0-\varepsilon} = 1] \tag{20}$$

that is, the ATE for the units for whom treatment changes discontinuously at z_0. If the policy is a small change in the threshold for program entry, the LATE parameter delivers the treatment effect for the subpopulation affected by the change, so that in that case it would be the parameter of policy interest.

The figure below illustrates the logic of regression discontinuity designs when treatment effects are heterogeneous across individuals. As before, identification relies on continuity of the potential outcome means $\mathbb{E}[Y_{0i} \mid Z_i = z]$ and $\mathbb{E}[Y_{1i} \mid Z_i = z]$ at the cutoff z_0, but now the gap between these two functions varies with the value of the running variable. This implies that treatment effects differ across values of Z_i, so that RD identification is inherently local.

In the sharp design, treatment status is fully determined by the cutoff rule, so that individuals with $Z_i < z_0$ are untreated and individuals with $Z_i \geq z_0$ are treated. As a result, the observed conditional expectation $\mathbb{E}[Y_i \mid Z_i = z]$ coincides



Heterogeneous effects: sharp RD

Heterogeneous effects: fuzzy RD

with $\mathbb{E}[Y_{0i} \mid Z_i = z]$ to the left of the cutoff and with $\mathbb{E}[Y_{1i} \mid Z_i = z]$ to the right. Under continuity, the discontinuity in $\mathbb{E}[Y_i \mid Z_i = z]$ at z_0 identifies the average treatment effect for individuals at the threshold, $\alpha_{RD} = \mathbb{E}[\alpha_i \mid Z_i = z_0]$. The figure also emphasizes that treatment effects away from the cutoff may differ from α_{RD}, highlighting that sharp RD does not justify extrapolation beyond z_0 when effects are heterogeneous.

In the fuzzy design, compliance with the cutoff rule is imperfect, so treated and untreated units coexist on both sides of the threshold. Consequently, the observed conditional mean $\mathbb{E}[Y_i \mid Z_i = z]$ is a mixture of potential outcomes on each side of z_0, and its discontinuity at the cutoff is not itself a treatment effect. Instead, the jump in outcomes reflects both the causal effect of treatment and the change in the probability of treatment at z_0. Identification therefore requires rescaling the outcome discontinuity by the corresponding discontinuity in $\mathbb{E}[D_i \mid Z_i = z]$, yielding a local Wald estimand. Depending on the assumptions imposed, this estimand can be interpreted either as an average treatment effect at the cutoff under local conditional independence, or as a local average treatment effect for compliers at the cutoff under a local monotonicity assumption.

V. Estimation Strategies

There are parametric and semiparametric estimation strategies. Hahn *et al.* (2001) suggested the following local estimator. Let $S_i \equiv \mathbb{1}\{z_0 - h < Z_i < z_0 + h\}$ where $h > 0$ denotes the bandwidth, and consider the subsample such that $S_i = 1$. The proposed estimator is the IV regression of Y_i on D_i using $W_i \equiv$

$\mathbb{1}\{z_0 < Z_i < z_0 + h\}$ as an instrument, applied to the subsample with $S_i = 1$:

$$\widehat{\alpha}_{RD} = \frac{\widehat{\mathbb{E}}[Y_i|W_i = 1, S_i = 1] - \widehat{\mathbb{E}}[Y_i|W_i = 0, S_i = 1]}{\widehat{\mathbb{E}}[D_i|W_i = 1, S_i = 1] - \widehat{\mathbb{E}}[D_i|W_i = 0, S_i = 1]}. \tag{21}$$

In sharp designs, the denominator is equal to 1. This estimator has nevertheless a poor boundary performance. An alternative is based on Equation (13). In the case of a sharp design, OLS provides consistent estimates of α_{RD}, but in the fuzzy design D_i is endogenous. In that context, we would typically use $\mathbb{1}\{Z_i \geq z_0\}$ as an instrument for D_i. These regression methods, not local to data points near the threshold, are implicitly predicated on the assumption of homogeneous treatment effects.

RD validity hinges on the absence of precise manipulation of the running variable around the cutoff z_0. If units can sort discontinuously around z_0 based on unobservables related to potential outcomes, the continuity condition in (2) may fail.

Common diagnostic checks include:

- Density test: the distribution of Z_i should be smooth at z_0 (a discontinuity may indicate sorting/manipulation).
- Covariate continuity: predetermined covariates should not display jumps at z_0; discontinuities suggest treated and control units differ systematically.
- Placebo cutoffs: re-estimate RD effects at thresholds where no treatment rule changes; significant "effects" cast doubt on the design.

Failure of these checks casts doubt on the fundamental RD assumption.

VI. Conditioning on Covariates

Even if the RD assumption is satisfied unconditionally, conditioning on covariates may mitigate the heterogeneity in treatment effects, hence contributing to the relevance of RD estimated parameters, which otherwise are "very local". Covariates may also make the local conditional exogeneity assumption more credible. Moreover, continuity of predetermined covariates at z_0 provides an important falsification test: covariates should not jump at the cutoff if units just below and above z_0 are comparable.

With heterogeneous treatment effects, RD identifies *local* causal effects at the cutoff z_0. Interpretation depends on the design and assumptions: in sharp RD, identification delivers an ATE at $Z_i = z_0$; in fuzzy RD, local conditional independence yields an ATE at $Z_i = z_0$, while local monotonicity yields a LATE for

compliers at $Z_i = z_0$. Extrapolation away from the cutoff generally requires additional assumptions and is not warranted by the RD design alone. Conditioning on covariates can sometimes reduce heterogeneity and improve external relevance.